

18/04/2025

CODE-A



Aakash

Medical | IIT-JEE | Foundations

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AIM - 720

(Advanced INTENSIVE Mastery for 720)

MM : 720

CST-7

Time : 180 Mins.

Complete Syllabus of NEET

Instructions:

- Duration of Test is 3 hrs.
- The Test consists of 180 questions. The maximum marks are 720.
- There are three parts in the question paper consisting of Physics, Chemistry and Biology having 45 questions each in Physics and Chemistry and 90 questions in Biology).
- Each question carries +4 marks. For every wrong response, -1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

6 Aakashians scored a perfect 720/720 from Classroom Programs

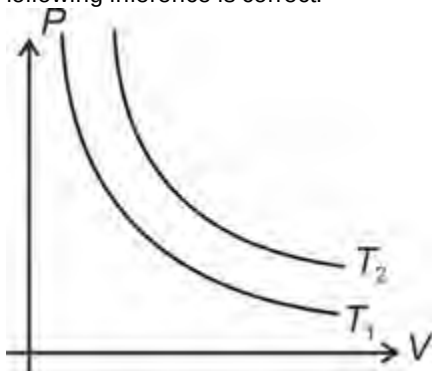


PHYSICS

1. A particle is moving in a circular path of radius ' R ' with constant speed ' v '. Magnitude of change in velocity of the particle when it moves through an angle of 180° , will be

(1) v
 (2) $\sqrt{2}v$
 (3) $2v$
 (4) Zero

2. Figure shows graph of pressure and volume of a gas sample at two temperatures T_1 and T_2 , which of the following inference is correct.



(1) $T_1 < T_2$
 (2) $T_1 > T_2$
 (3) $T_1 = T_2$
 (4) Data insufficient

3. Consider the following statements.
 (A) An ideal gas expands in an adiabatic chamber. The change in internal energy of gas should be zero.
 (B) Efficiency of a heat engine can't be greater than the Carnot engine working between same temperature limits.
 (C) Heat and work are considered to be point functions.
 Based on above information pick the correct option.

(1) Only statement (B) is correct.
 (2) Only statements (B) and (C) are correct.
 (3) Only statements (A) and (C) are correct.
 (4) All statements are correct.

4. The excess pressure inside a soap bubble A is twice that in another soap bubble B. The ratio of volumes of A and B is

(1) 4 : 1
 (2) 1 : 8
 (3) 1 : 2
 (4) 1 : 16

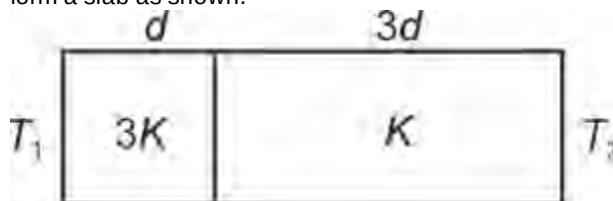
5. Consider the following statements.

(A) Fluids offer significant resistance to change in their shape.
 (B) The contact angle between water and glass is acute.
 (C) A thin needle placed gently on surface of water can float.

Based on above information pick the correct option.

(1) Both statements (A) and (B) are correct.
 (2) Both statements (B) and (C) are correct.
 (3) Both statements (A) and (C) are correct.
 (4) All statements are correct.

6. Two materials having coefficient of thermal conductivities as $3K$ and K and lengths d and $3d$ respectively are joined to form a slab as shown.



The equivalent thermal conductivity of composite slab will be (Area of cross-section is same for both the materials)

(1) $4K$
 (2) $\frac{6K}{5}$
 (3) $\frac{5K}{6}$
 (4) $2K$

7. The average velocity of gas molecules having molar mass M at temperature T will be (R is universal gas constant)

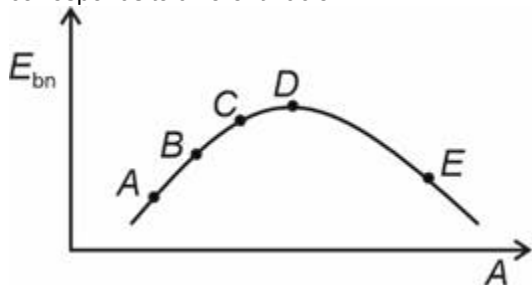
(1) $\sqrt{\frac{3RT}{M}}$
 (2) $\sqrt{\frac{8RT}{M}}$
 (3) $\sqrt{\frac{2RT}{M}}$
 (4) Zero

8. If E_1 is the energy of the 1st Bohr's orbit and E_2 is the energy of 2nd Bohr's orbit, a transition between them would result in the emission of radiation of frequency (h is Planck's constant)

(1) $E_2 - E_1$
 (2) $E_1 - E_2$
 (3) $\frac{E_2 - E_1}{h}$
 (4) $\frac{E_1 - E_2}{h}$

9. The number of photons of wavelength 450 nm emitted per second by an electric bulb of power 200 W nearly is (Take $h = 6 \times 10^{-34}$ J s)
- 100
 - 10^{20}
 - 5×10^{20}
 - 5×10^{24}

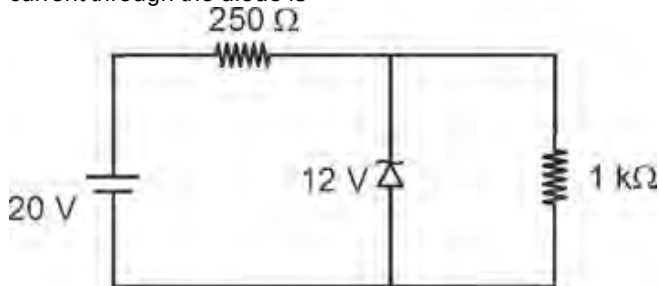
10. The graph of binding energy per nucleon E_{bn} versus nucleus mass number A is given below. A, B, C, D and E corresponds to different nuclei.



In which of the following reactions energy may release?

- (i) $A + B \rightarrow C$ (ii) $B + C \rightarrow D$
 (iii) $E \rightarrow D + C$ (iv) $D + C \rightarrow E$

- (i) and (ii) only
 - (i) and (iii) only
 - (i) and (iv) only
 - (i), (ii) and (iii)
11. In a semiconductor diode, the barrier potential offers opposition to
- Free electrons in p -region
 - Holes in n -region
 - Majority carriers in both regions
 - Majority as well as minority carriers in both regions.
12. A Zener diode, having breakdown voltage equal to 12 V, is used in a voltage regulator circuit as shown in figure. The current through the diode is



- 10 mA
- 12 mA
- 20 mA
- 32 mA

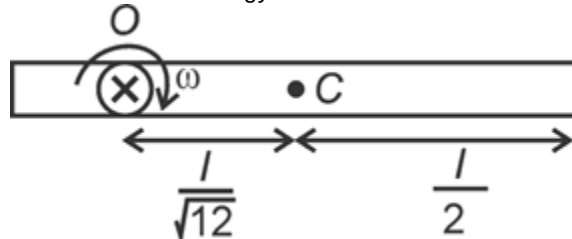
13. The amplitude of oscillations is reduced to half of its initial value of 24 cm (due to damping by resistive forces) at the end of 20 oscillations. After it completes 60 oscillations its amplitude will be

- 6 cm
- 12 cm
- 3 cm
- 18 cm

14. A torque of $(4t)$ N m is applied on a wheel having angular momentum of $4 \text{ kg m}^2 \text{ s}^{-1}$. Angular momentum of the wheel after 4 seconds is (in $\text{kg m}^2 \text{ s}^{-1}$)

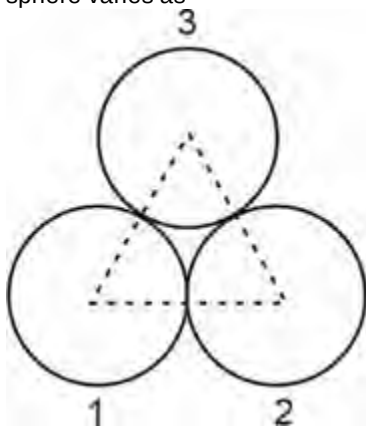
- 32
- 36
- 64
- 68

15. A uniform rod of length l and mass m is hinged at point O as shown. If it rotates with n rotations per second, then rotational kinetic energy of the rod is



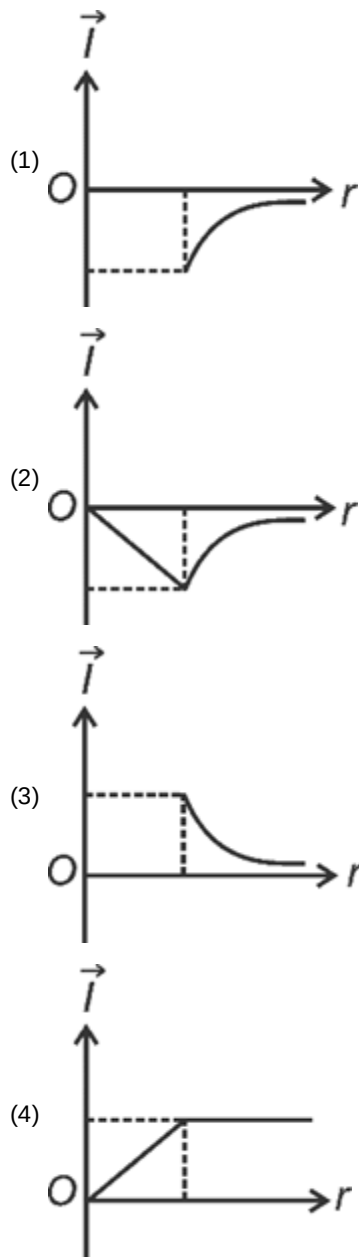
- $\frac{ml^2 \pi^2 n^2}{6}$
- $\frac{ml^2 \pi^2 n^2}{3}$
- $\frac{ml^2 \pi^2 n^2}{12}$
- $\frac{2ml^2 \pi^2 n^2}{3}$

16. Three solid spheres of same size (radius R) and made up of same material are placed in contact with each other as shown below. Resultant gravitational force (F) acting on any sphere varies as



- (1) $F \propto \frac{1}{R^2}$
 (2) $F \propto R^0$
 (3) $F \propto R^2$
 (4) $F \propto R^4$

17. The correct graph depicting variation of gravitational field intensity ($\vec{I}$) due to solid sphere (having uniform density) with distance r from its centre, is



18. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : The power of a double convex lens ($\mu = \frac{3}{2}$) immersed in water ($\mu = \frac{4}{3}$) is $\frac{1}{3}$ times the power in air.

Reason (R) : Focal length of the above lens becomes 3 times its focal length in air.

In the light of the above statements, choose the **correct** answer from the options given below :

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 (3) (A) is true but (R) is false
 (4) Both (A) and (R) are false
19. When an object is at a distance 30 cm and 50 cm from a lens, virtual image and a real image is formed respectively having same magnification. The focal length of the lens is
- (1) 130 cm
 (2) 20 cm
 (3) 40 cm
 (4) 80 cm

20. A string fixed at both ends oscillates in 5 segments. If length of string is 10 m and velocity of wave is 20 m/s, then the frequency of the wave is

- (1) 10 Hz
 (2) 5 Hz
 (3) 15 Hz
 (4) 20 Hz

21. Which of the following relation between acceleration and displacement represents an SHM (where a is acceleration and x is displacement)

- (1) $a = 2x$
 (2) $a = 2x^2$
 (3) $a = -2x$
 (4) $a = -2x^2$

22. The bulk modulus of a spherical object is ' B '. If it is subjected to uniform pressure P , the fractional decrease in radius of the object is

- (1) $\frac{B}{3P}$
 (2) $\frac{3P}{B}$
 (3) $\frac{P}{3B}$
 (4) $\frac{P}{B}$

23. The electric field and magnetic field associated with an EM wave, propagating along the $+z$ axis, can be represented by

- (1) $\vec{E} = E(z, t)\hat{i}$, $\vec{B} = B(z, t)\hat{j}$
 (2) $\vec{E} = E(z, t)\hat{k}$, $\vec{B} = B(z, t)\hat{i}$
 (3) $\vec{E} = E(z, t)\hat{j}$, $\vec{B} = B(z, t)\hat{i}$
 (4) $\vec{E} = E(z, t)\hat{j}$, $\vec{B} = B(z, t)\hat{k}$

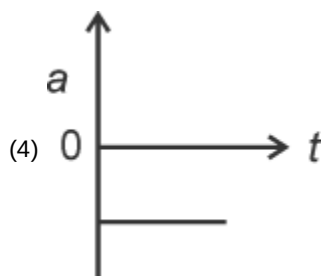
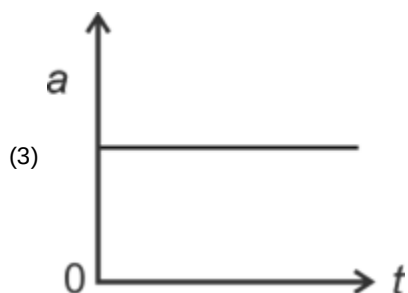
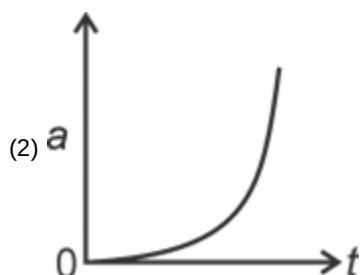
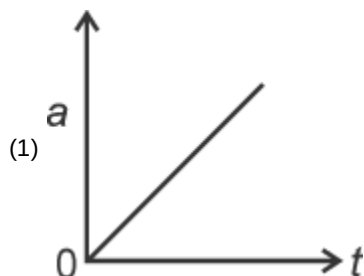
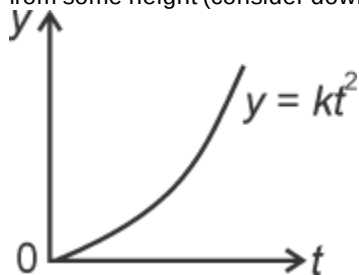
24. Match List-I with List-II.

	List-I		List-II
a.	All bright fringes are of equal width are observed in	(i)	Malus law
b.	Decreasing intensity of secondary bright fringes are observed in	(ii)	YDSE
c.	Intensity of light passing through analyser is given by	(iii)	Huygen's principle
d.	Concept of secondary wavelets was introduced by	(iv)	Diffraction

Choose the correct answer from the options given below.

- (1) a(iii), b(i), c(ii), d(iv)
 (2) a(ii), b(iv), c(i), d(iii)
 (3) a(ii), b(i), c(iii), d(iv)
 (4) a(i), b(ii), c(iii), d(iv)
25. If the random error in the arithmetic mean of 100 observations is x , then the random error in the arithmetic mean of 500 observations will be
- (1) $5x$
 (2) $\frac{x}{5}$
 (3) x
 (4) $\frac{x}{25}$
26. If a steel sphere is heated, then the maximum percentage change will be observed in its
- (1) Radius
 (2) Volume
 (3) Area
 (4) Mass

27. Choose the correct representation of acceleration-time graph for the given position-time curve of a body dropped from some height (consider downward direction as positive)



28. Consider the given statements and choose the correct option for motion in one dimension.

Statement I: If a particle is speeding up, acceleration is in the direction of velocity.

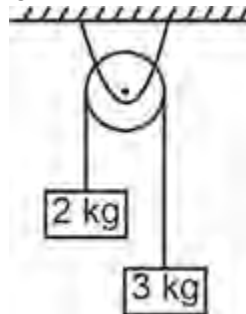
Statement II: If a particle is speeding down, acceleration is in the direction opposite to that of velocity.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Only statement A is true
 - (2) Only statement B is true
 - (3) Both statements A and B are true
 - (4) Both statements A and B are false
29. A spring of length L and spring constant K is cut into two parts of length $\frac{L}{3}$ and $\frac{2L}{3}$, then the spring constant of each part respectively will be

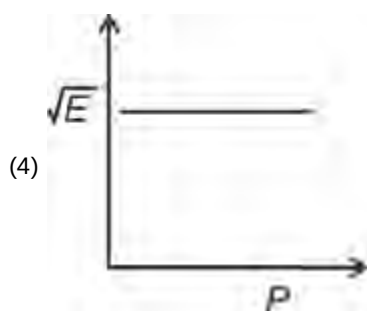
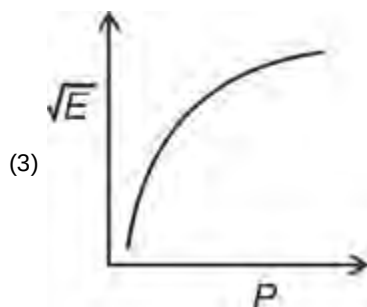
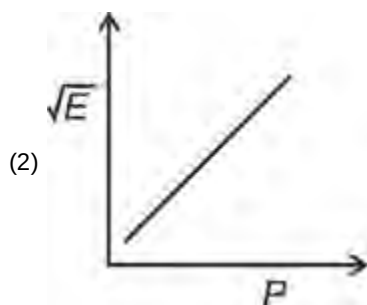
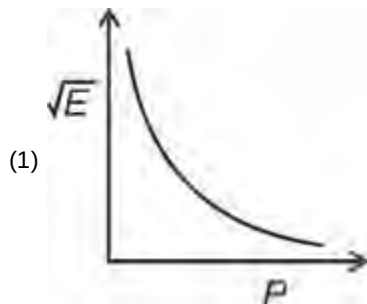
- (1) $\frac{K}{2}$ and $\frac{K}{3}$
- (2) $3K$ and $\frac{3K}{2}$
- (3) $\frac{K}{3}$ and $\frac{K}{4}$
- (4) K and K

30. Two bodies are connected with a string and the string passes over an atwood's machine. The tension in the string is



- (1) 12 N
- (2) 24 N
- (3) 30 N
- (4) 50 N

31. Choose the correct representation of $\sqrt{E}$ vs P where E is kinetic energy and P is linear momentum of a body.



32. A 0.1 kg ball makes an elastic head-on collision with a ball of unknown mass which is initially at rest. If the ball rebounds with one-third of its original speed, the mass of other ball is

- (1) 0.1 kg
- (2) 0.2 kg
- (3) 0.3 kg
- (4) 0.4 kg

33. The magnetic flux linked with a circular coil having resistance $4\ \Omega$ is $\phi = (2t^2 + 4t + 4)$ Wb. The current flowing through the coil at $t = 2$ s will be

- (1) 3 A
- (2) 1 A
- (3) 1.5 A
- (4) 2 A

34. The self-inductance of the solenoid is L . If number of turns per unit length is doubled keeping its length same, then new inductance of the solenoid will become

- (1) 2 times
- (2) 4 times
- (3) $\frac{1}{2}$ times
- (4) $\frac{1}{4}$ times

35. Two similar coils of radius R are lying concentrically with their planes at right angle to each other. The current flowing in them are $3I$ and $4I$, respectively, the resultant magnetic field induction at the centre will be

- (1) $\frac{3I\mu_0}{2R}$
- (2) $\frac{7I\mu_0}{2R}$
- (3) $\frac{5I\mu_0}{2R}$
- (4) Zero

36. A charged particle of charge q is moving with velocity $\vec{v}$ in uniform magnetic field $\vec{B}$. If force acting on the particle is $\vec{F}$, then

- (1) $\vec{F} \times \vec{v} = \vec{0}$
- (2) $\vec{F} \cdot \vec{B} = 0$
- (3) $\vec{F} \cdot \vec{v} = 0$
- (4) Both (2) and (3)

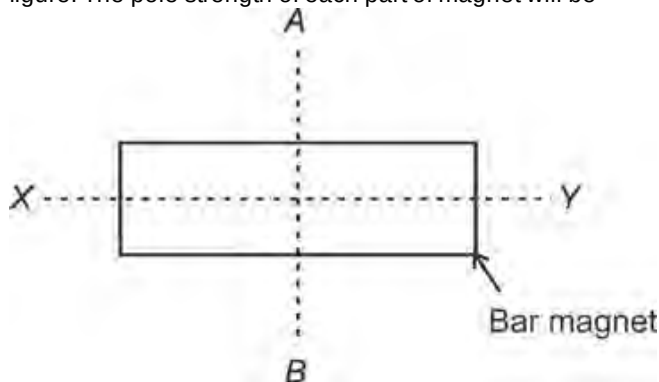
37. The potential difference across the resistance, capacitance and inductance are 80 V, 40 V and 100 V respectively in series LCR circuit. The phase angle between current and source voltage in the circuit will be

- (1) 45°
- (2) 53°
- (3) 37°
- (4) 90°

38. An inductor of inductance L has reactance X . If inductance and frequency of AC source both are doubled, then the new reactance will be

(1) $2X$
 (2) $\frac{X}{2}$
 (3) $4X$
 (4) $\frac{X}{4}$

39. A bar magnet of magnetic moment M and pole strength m is cut into four equal parts along AB and XY as shown in the figure. The pole strength of each part of magnet will be



(1) m
 (2) $4m$
 (3) $\frac{m}{2}$
 (4) $\frac{m}{4}$

40. Two charged conducting spheres of radii a and b are connected through a wire. The ratio of their surface charge density at equilibrium will be

(1) $\frac{b^2}{a^2}$
 (2) $\sqrt{\frac{b}{a}}$
 (3) $\left(\frac{b}{a}\right)^{3/2}$
 (4) $\frac{b}{a}$

41. Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

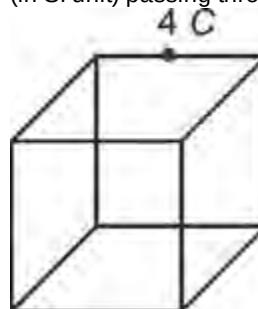
Assertion (A) : In an isolated system, if positive charge is destroyed somehow, then equal amount of negative charge will be destroyed.

Reason (R) : Total charge of an isolated system remain conserved.

In the light of the above statements, choose the **correct** answer from the options given below :

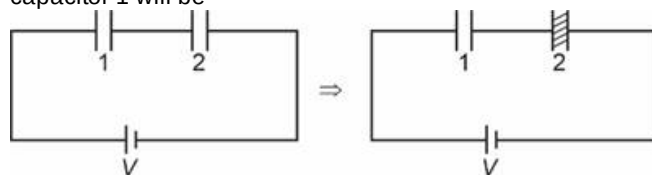
(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 (3) (A) is true but (R) is false
 (4) (A) is false but (R) is true

42. A point charge of 4 C is placed at the midpoint of an edge of a cubical gaussian surface as shown in figure. The total flux (in SI unit) passing through the cube is



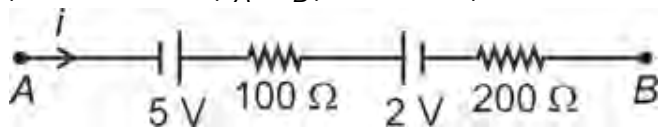
(1) $\frac{4}{\epsilon_0}$
 (2) $\frac{8}{\epsilon_0}$
 (3) $\frac{1}{\epsilon_0}$
 (4) $\frac{2}{\epsilon_0}$

43. Two identical air capacitors are connected in series. When they are connected with a battery the final charge on capacitor 1 is Q_0 . Afterwards, a dielectric of dielectric constant 2 is inserted into the 2nd capacitor while the system remain connected to the battery. The final charge on capacitor 1 will be

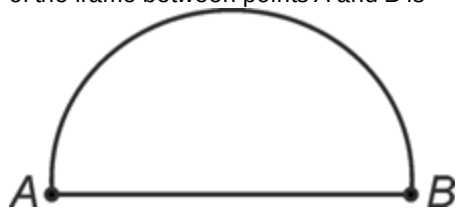


(1) Q_0
 (2) $\frac{4Q_0}{3}$
 (3) $\frac{2Q_0}{3}$
 (4) $\frac{8Q_0}{3}$

44. A variable current $i = (2t + 1)\text{mA}$ is flowing through a part of a circuit as shown below. The potential difference between points A and B i.e. $(V_A - V_B)$ at $t = 2\text{ s}$ is equal to



- (1) 1.5 V
 (2) -3 V
 (3) 3 V
 (4) -1.5 V
45. A semicircular wire frame of radius r is formed. If resistance per unit length of wire is R_0 , then the equivalent resistance of the frame between points A and B is



- (1) $\frac{2\pi R_0 r}{2 + \pi}$
 (2) $\frac{\pi R_0 r}{1 + \pi}$
 (3) $\frac{(2\pi + r)R_0}{2 + \pi}$
 (4) $\frac{2(\pi + r)R_0}{1 + \pi}$

CHEMISTRY

46. Which among the following ions gives a gas with rotten egg like smell on reaction with dilute H_2SO_4 ?

- (1) NO_3^-
- (2) CO_3^{2-}
- (3) Cl^-
- (4) S^{2-}

47. Given below are two statements:

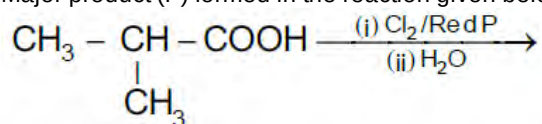
Statement I: White phosphorus is a translucent white waxy solid.

Statement II: P–P–P bond angle in white phosphorus is 90° .

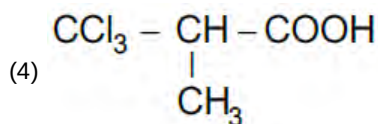
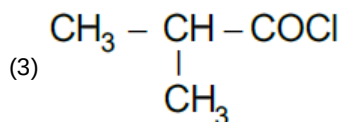
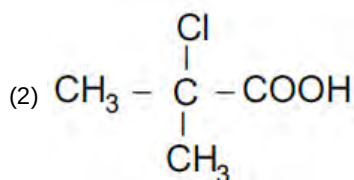
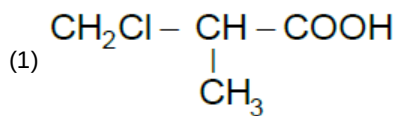
In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

48. Major product (P) formed in the reaction given below is



(P) (Major)



49. Which among the following will not undergo Cannizzaro reaction in presence of concentrated alkali?

- (1) HCHO
- (2) $\text{C}_6\text{H}_5\text{CHO}$
- (3) $(\text{CH}_3)_3\text{CCHO}$
- (4) $\text{CH}_3 - \underset{\text{O}}{\overset{\parallel}{\text{C}}} - \text{CH}_3$

50. Half life of a reaction with rate constant $1.386 \times 10^{-3} \text{ s}^{-1}$ will be

- (1) $7.8 \times 10^2 \text{ s}$
- (2) 500 s
- (3) 250 s
- (4) 1000 s

51. Given below are two statements

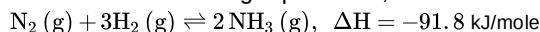
Statement I: Catalyst increases rate of reaction by decreasing activation energy.

Statement II: A catalyst can catalyse both spontaneous and non-spontaneous reactions.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

52. Consider the following equilibrium,



Formation of NH_3 is favoured by,

- (a) Increasing the concentration of $\text{H}_2(\text{g})$.
- (b) Increasing pressure.
- (c) Decreasing temperature

Choose the correct option

- (1) (a) and (c) only
- (2) (a), (b) and (c)
- (3) (a) and (b) only
- (4) (b) and (c) only

53. Which among the following equilibrium has $K_p = K_c$?

- (1) $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$
- (2) $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
- (3) $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$
- (4) $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$

54. Given below are two statements:
Statement I: As_2O_3 is amphoteric in nature.
Statement II: NO_2 is a neutral oxide of nitrogen
 In the light of the above statements, choose the most appropriate option.
 (1) Statement I is correct but statement II is incorrect
 (2) Statement I is incorrect but statement II is correct
 (3) Both statement I and statement II are correct
 (4) Both statement I and statement II are incorrect
55. An element has electronic configuration as $[\text{Ar}] 3d^7 4s^2$. Its group number in modern periodic table is.
 (1) 6
 (2) 9
 (3) 8
 (4) 10
56. If the standard reduction potential for $\text{Fe}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Fe}(\text{s})$ and $\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Fe}^{2+}(\text{aq})$ are x volt and y volt respectively then the value of E° for Fe^{2+}/Fe will be
 (1) $\frac{3y-x}{2} \text{V}$
 (2) $\frac{3x-y}{2} \text{V}$
 (3) $\frac{y-3x}{2} \text{V}$
 (4) $\frac{x-3y}{2} \text{V}$
57. Correct order of limiting molar conductivity of given ions at 298 K in water is
 (1) $\text{SO}_4^{2-} > \text{OH}^- > \text{Br}^- > \text{Cl}^-$
 (2) $\text{Br}^- > \text{Cl}^- > \text{OH}^- > \text{SO}_4^{2-}$
 (3) $\text{OH}^- > \text{SO}_4^{2-} > \text{Br}^- > \text{Cl}^-$
 (4) $\text{OH}^- > \text{Br}^- > \text{Cl}^- > \text{SO}_4^{2-}$
58. 2 g of non-electrolyte solute dissolved in 100 g benzene, lowered the freezing point of solution by 0.50 K. If molal depression constant of benzene is $5.12 \text{ K Kg mol}^{-1}$ then molar mass of the solute (in g mol^{-1}) will be
 (1) 224.8
 (2) 204.8
 (3) 102.4
 (4) 208.4
59. Consider the following binary solutions
 (a) Ethanol and acetone
 (b) Benzene and toluene
 (c) Phenol and aniline
 (d) Carbon disulphide and acetone
 The solutions which show positive deviation from Raoult's law are
 (1) (a) and (b) only
 (2) (b) and (c) only
 (3) (a), (c) and (d) only
 (4) (a) and (d) only
60. During the conversion of O_2 to O_2^+ ion the electron is removed from which of the following molecular orbitals?
 (1) π^*
 (2) π
 (3) σ
 (4) σ^*
61. Given below are the two statements one is labelled as assertion (A) the other is labelled as reason (R).
Assertion (A): Boiling point of HF is higher than HCl.
Reason (R): HF forms intermolecular hydrogen bond with each other.
 In the light of above statements, choose the **correct** answer.
 (1) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 (2) (A) is true but (R) is false
 (3) (A) is false but (R) is true
 (4) Both (A) and (R) are true, (R) is the correct explanation of (A)
62. The correct order of bond angle of the given compounds is
 (1) $\text{H}_2\text{O} < \text{CH}_4 < \text{NH}_3$
 (2) $\text{NH}_3 < \text{CH}_4 < \text{H}_2\text{O}$
 (3) $\text{H}_2\text{O} < \text{NH}_3 < \text{CH}_4$
 (4) $\text{CH}_4 < \text{NH}_3 < \text{H}_2\text{O}$
63. The coordination number of cobalt in $[\text{CoCl}_2(\text{en})_2]^+$ is
 (1) 8
 (2) 6
 (3) 5
 (4) 4

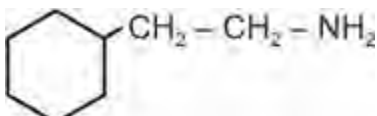
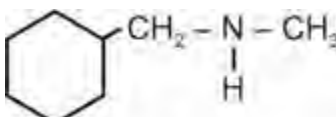
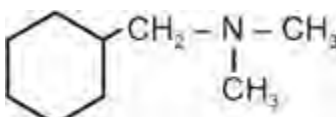
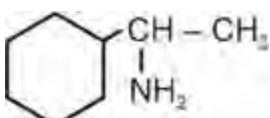
64. Match List-I with List-II

	List-I Complex		List-II Crystal field stabilisation energy (ignoring pairing energy)
a.	$[\text{Fe}(\text{CN})_6]^{3-}$	(i)	$-2.4\Delta_0$
b.	$[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$	(ii)	$-0.4\Delta_0$
c.	$[\text{MnCl}_6]^{3-}$	(iii)	$-0.6\Delta_0$
d.	$[\text{CoF}_6]^{3-}$	(iv)	$-2.0\Delta_0$

The correct match is

- (1) a(ii), b(iii), c(i) d(iv)
 (2) a(i), b(iv), c(ii) d(iii)
 (3) a(iii), b(iv), c(ii) d(i)
 (4) a(iv), b(i), c(iii) d(ii)

65. Which of the following will not react with Hinsberg's reagent?

- (1) 
 (2) 
 (3) 
 (4) 

66. Given below are two statements

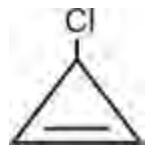
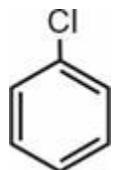
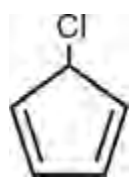
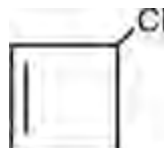
Statement I: Phenylmethanamine cannot be prepared through Gabriel phthalimide synthesis.

Statement II: Phenylmethanamine reacts with nitrous acid and liberates nitrogen gas.

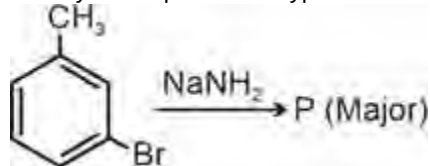
In the light of the above statements, choose the most appropriate answer from the options given below.

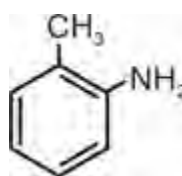
- (1) Both statement I and statement II are correct
 (2) Both statement I and statement II are incorrect
 (3) Statement I is correct but statement II is incorrect
 (4) Statement I is incorrect but statement II is correct

 67. Which one of the following gives fastest reaction with aqueous AgNO_3 solution?

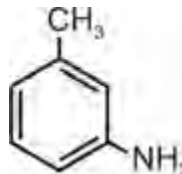
- (1) 
 (2) 
 (3) 
 (4) 

68. Identify P and predict the type of reaction

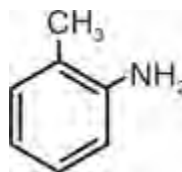


- (1) 

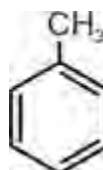
and substitution reaction

- (2) 

and substitution reaction

- (3) 

and elimination addition reaction

- (4) 

and elimination reaction

69. Which of the following oxoacid of sulphur has peroxy (–O–O–) linkage?

- (1) Sulphurous acid
- (2) Pyrosulphuric acid
- (3) Sulphuric acid
- (4) Peroxodisulphuric acid

70. Consider the following statements.

- (a) Chlorine is greenish yellow gas with pungent and suffocating odour
- (b) Chlorine on reaction with dry slaked lime gives bleaching powder.
- (c) Chlorine water on standing loses its yellow colour due to the formation of HCl and HOCl

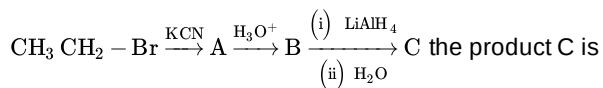
The correct statements are

- (1) (a) and (b) only
- (2) (b) and (c) only
- (3) (a), (b) and (c)
- (4) (a) and (c) only

71. The product which is not formed on the complete hydrolysis of XeF_4 is

- (1) HF
- (2) O_2
- (3) Xe
- (4) F_2

72. Consider the following sequence of reactions,

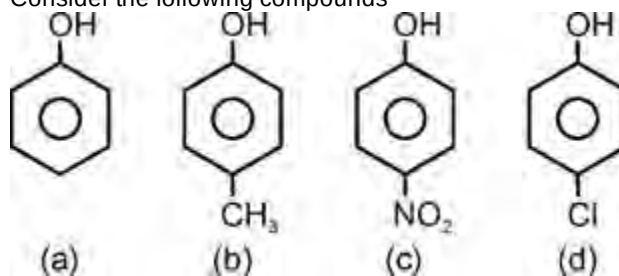


- (1) Ethanoic acid
- (2) n-propylalcohol
- (3) Ethyl alcohol
- (4) Propanal

73. When vapours of a tertiary alcohol is passed over heated copper at 573 K, the major product formed is

- (1) An aldehyde
- (2) A ketone
- (3) An alkene
- (4) A carboxylic acid

74. Consider the following compounds



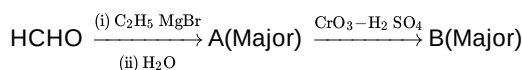
The decreasing order of acidic strength of the above compounds is:

- (1) $c > d > a > b$
- (2) $d > c > a > b$
- (3) $a > b > c > d$
- (4) $b > d > c > a$

75. Which of the following compounds is most difficult to protonate?

- (1) Phenol
- (2) Water
- (3) Methanol
- (4) Methoxymethane

76. Consider the following reaction sequence



Major product B

- (1) $\text{CH}_3\text{CH}_2\text{COOH}$
- (2) $\text{CH}_3\text{CH}_2\text{CHO}$
- (3) $\text{CH}_3\text{CH}=\text{CH}_2$
- (4) CH_3COCH_3

77. Match the List-I with List-II

	List-I (Compound)		List-II (Heteroatom in ring)
a.	Pyridine	(i)	No heteroatom
b.	Furan	(ii)	Sulphur
c.	Thiophene	(iii)	Oxygen
d.	Tropone	(iv)	Nitrogen

Choose the correct match from the options given below.

- (1) a(iii), b(i), c(ii), d(iv)
- (2) a(iii), b(iv), c(ii), d(i)
- (3) a(iv), b(iii), c(i), d(ii)
- (4) a(iv), b(iii), c(ii), d(i)

78. Consider the following statements:

Statement I: Oxidation state of Cr in both CrO_5 and $\text{K}_2\text{Cr}_2\text{O}_7$ is same

Statement II: S₈ on disproportionation reaction in alkaline medium gives sulphide and sulphate ions.
In the light of above statements, choose the correct option.

- (1) Both statements I and statement II are incorrect
- (2) Statement I is incorrect but statement II is correct
- (3) Both statements I and statement II are correct
- (4) Statement I is correct but statement II is incorrect

79. The stability of +1 oxidation state increases in the sequence of

- (1) $\text{In} < \text{Tl} < \text{Ga} < \text{Al}$
- (2) $\text{Tl} < \text{In} < \text{Al} < \text{Ga}$
- (3) $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$
- (4) $\text{Al} < \text{In} < \text{Ga} < \text{Tl}$

80. Consider the following statements

- (a) α -form and β -form of glucose are called anomers.
- (b) α -D-(+)-glucopyranose has five chiral centres.
- (c) melting point of α -form of glucose and β -form of glucose is same.

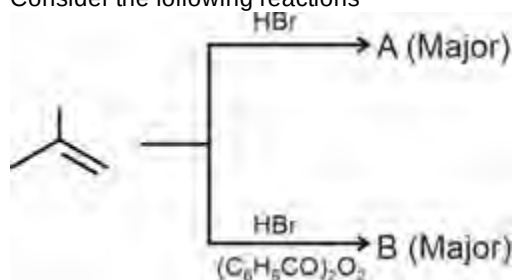
The correct statements are

- (1) (a) and (b) only
- (2) (b) and (c) only
- (3) (a) and (c) only
- (4) (a), (b) and (c)

81. Maximum number of electrons in a subshell of an atom is given by

- (1) $4l$
- (2) $2(2l + 1)$
- (3) $2l + 1$
- (4) $2(l^2 - 1)$

82. Consider the following reactions



Major products (A) and (B) respectively are

- (1) and
- (2) and
- (3) and
- (4) and

83. Given below are the two statements

Statement I: Protein found in a biological system with a unique three-dimensional structure and biological activity is called a native protein.

Statement II: During denaturation of protein secondary, tertiary and primary structures are destroyed.

In light of above statements, choose the correct answer.

- (1) Statement I is correct but statement II is incorrect
- (2) Both statement I and statement II are correct
- (3) Both statement I and statement II are incorrect
- (4) Statement I is incorrect but statement II is correct

84. Consider the following statements

- (a) Canal rays are positively charged gaseous ions.
- (b) Cathode rays consist of negatively charged particles called electrons.

(c) $\frac{e}{m}$ value of electron is $1.758820 \times 10^{11} \text{ C kg}^{-1}$

The correct statements are

- (1) (a) and (b) only
- (2) (b) and (c) only
- (3) (a) and (c) only
- (4) (a), (b) and (c)

85. Total number of structural isomers of C_7H_{16} is

- (1) 7
- (2) 8
- (3) 9
- (4) 10

86. Conc. H_2SO_4 is 98% H_2SO_4 by mass and has a density of solution 1.84 g mL^{-1} . What volume of the conc. acid is required to make 2.5 litre of 0.5 M H_2SO_4 solution?

(1) 68 mL
 (2) 13.6 mL
 (3) 68 L
 (4) 13.6 L

87. Given below are two statements.

Statement I: In acidic solution, yellow coloured chromate ions are converted to orange coloured dichromate ions.

Statement II: Manganate ions disproportionate in a neutral or acidic solution to give permanganate and manganese dioxide.

In the light of above statements, choose the **correct** answer.

(1) Both statement I and statement II are correct
 (2) Both statement I and statement II are incorrect
 (3) Statement I is correct but statement II is incorrect
 (4) Statement I is incorrect but statement II is correct

88. Match the ions in List-I with their spin only magnetic moment value in Bohr magneton in List-II.

	List-I		List-II
a.	Fe^{3+}	(i)	3.87
b.	V^{2+}	(ii)	0
c.	Cu^+	(iii)	5.92
d.	Co^{3+}	(iv)	4.89

Choose the correct answer from the options given below:

(1) a(i), b(ii), c(iv), d(iii)
 (2) a(iii), b(i), c(ii), d(iv)
 (3) a(iii), b(ii), c(i), d(iv)
 (4) a(i), b(iii), c(ii), d(iv)

89. Group reagent for Ba^{2+} , Sr^{2+} and Ca^{2+} is

(1) H_2S is presence of dil.HCl
 (2) NH_4OH in presence of NH_4Cl
 (3) $(\text{NH}_4)_2\text{CO}_3$ in presence of NH_4OH
 (4) H_2S is presence of NH_4OH

90. Consider the following processes.

	ΔH
$\text{C}(\text{graphite}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$	$-x \text{ kJ}$
$2\text{CO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g})$	$-y \text{ kJ}$

For $\text{C}_{(\text{graphite})} + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{CO}(\text{g})$, ΔH in kJ will be

(1) $\frac{2y-x}{2}$
 (2) $\frac{y-x}{2}$
 (3) $\frac{2y+x}{2}$
 (4) $\frac{y-2x}{2}$

BIOLOGY

91. Which of the following has water-containing cavities within the vascular bundles, but they are devoid of pith and pericycle?

- (1) Monocot root
- (2) Monocot stem
- (3) Dicot stem
- (4) Dicot leaf

92. Match the column I with column II and select the **correct** option.

	Column I		Column II
(a)	Vessel	(i)	Commonly found in pulp of pear and sapota
(b)	Phloem parenchyma	(ii)	Found in layers below epidermis in most of the dicots
(c)	Sclereids	(iii)	Made up of elongated, tapering cylindrical cells having dense cytoplasm and nucleus
(d)	Collenchyma	(iv)	Members are interconnected through perforations in their common walls

- (1) a(ii), b(iv), c(iii), d(i)
- (2) a(iii), b(ii), c(i), d(iv)
- (3) a(iv), b(iii), c(i), d(ii)
- (4) a(iv), b(ii), c(iii), d(i)

93. In a given sample of RNA, there are 666 bases, that code for a protein with 222 amino acids and the base at position 601 is deleted such that length of RNA now becomes 665 base long. How many codons will be altered?

- (1) 220
- (2) 22
- (3) 222
- (4) 11

94. The enzyme that uses a template to catalyse the polymerisation of deoxynucleotides is

- (1) Topoisomerase
- (2) Polynucleotide phosphorylase
- (3) RNA polymerase II
- (4) DNA polymerase

95. In *Escherichia coli*,

- (a) genes are split
 - (b) intervening sequences appear in mRNA
 - (c) structural gene is polycistronic
 - (d) different types of proteins are synthesised by monocistronic gene
- Choose the **correct** statement(s).

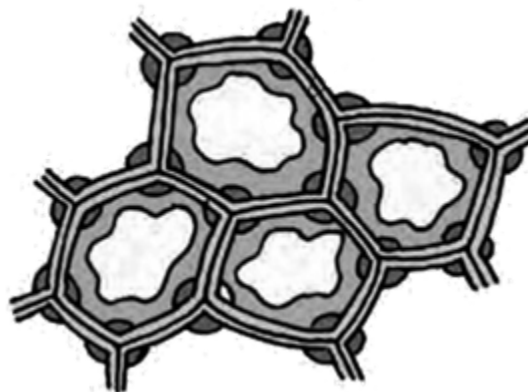
- (1) (a) and (c) only
- (2) (a), (b) and (d) only
- (3) All except (d)
- (4) (c) only

96. In the below given sequences of mRNA, which will form the translation unit to form polypeptide having maximum number of amino acids?

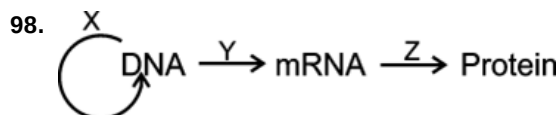
- (a) 3'-AAUCGUAUUCACGACGUA-5'
 - (b) 5'-CCUUGGUCAGGGAAG-3'
 - (c) 3'-AUUGUGGGCGUUCGUGUA-5'
 - (d) 5'-AUGAUCUAUUCAGAAUGA-3'
- The **correct** one(s) is/are

- (1) (b) and (c) only
- (2) (a) and (d) only
- (3) All except (b)
- (4) (d) only

97. Observe the figure given below and choose the option having the **most appropriate** explanation.



- (1) This type of tissue is found in both monocotyledonous and dicotyledonous plants
This type of tissue resists the bending of the stem and pulling out and tearing of young leaves due to action of wind
- (2) At the corners of these cells, suberin and lignin depositions are found
- (3) This type of tissue forms the major component within organs of plants



In the given central dogma in molecular biology, **correctly** fill in the blanks for X, Y and Z.

- (1) X → translation, Y → replication, Z → transcription
- (2) X → reverse transcription, Y → replication, Z → translation
- (3) X → replication, Y → transcription, Z → translation
- (4) X → transcription, Y → translation, Z → replication

99. Some bacteria have a loose sheath called ———, which protects the cell from loss of water and nutrients. Choose the **correct** option to fill in the blank.

- (1) Capsule
- (2) Slime layer
- (3) Peptidoglycan
- (4) Cell wall

100. "9 + 0" organization of microtubules is observed in

- (1) Flagella of bacteria
- (2) Cilia of eukaryotes
- (3) Basal body of eukaryotic flagella
- (4) Microfilaments

101. Read the following statements of Assertion (A) and Reason (R).

Assertion (A): Lysosomes contain enzymes for the digestion of macromolecules.

Reason (R): Hydrolytic enzymes present in the lysosomes are active at basic pH.

In light of the above statements, choose the **correct** answer from the following options.

- (1) Both (A) and (R) are true, and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true, but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) (A) is false but (R) is true

102. Which of the following events takes place in the prophase of mitosis?

- (1) Chromosomes align themselves at the equator
- (2) The nuclear envelope begins to break down and the nucleolus disappears
- (3) Chromosome duplication
- (4) Chromatids are separated and move to opposite poles.

103. Find the **incorrect** match.

(1) Plasmid	–	Extrachromosomal DNA
(2) Lysosomes	–	Formed by packaging in Golgi body
(3) Golgi body	–	Cis face is maturing face
(4) Chromoplasts	–	Coloured plastids

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

104. Read the following statements and arrange the corresponding events in the **correct** chronological sequence w.r.t meiosis.

- (a) The nucleolus and nuclear membrane reappear
 - (b) Chiasmata formation
 - (c) Exchange of genetic material between non-sister chromatids of the homologous chromosomes
 - (d) Nucleolus disappears and the nuclear envelope disintegrate
 - (e) Splitting of the centromere holding the two chromatids
- The **correct** sequence of events is

- (1) (a) → (c) → (d) → (b) → (e)
- (2) (c) → (b) → (d) → (e) → (a)
- (3) (c) → (b) → (d) → (a) → (e)
- (4) (c) → (d) → (b) → (e) → (a)

105. Which of the following options has taxonomic categories showing hierarchical arrangement in an ascending order?

- (1) Species → Genus → Family → Order → Class
- (2) Class → Order → Family → Genus → Species
- (3) Family → Order → Class → Species → Genus
- (4) Order → Class → Genus → Family → Species

106. Consider the given features.

- (a) Possesses compactly packed fibres and fibroblasts
 - (b) Attaches skeletal muscles to bones
 - (c) Orientation of fibres show a regular pattern
- Choose the option for which all of the above given features hold true.

- (1) Tendon
- (2) Ligament
- (3) Adipose tissue
- (4) Areolar tissue

107. Complete the analogy by selecting the correct option.

Glycerol: Three hydroxyl groups :: Glucose: _____ hydroxyl groups

- (1) Four
- (2) Six
- (3) Five
- (4) Three

108. Select the **incorrect** statement.

- (1) GLUT-4 is a heteropolymer.
- (2) Paper made from plant pulp and cotton fibre is majorly collagenous in nature.
- (3) Dietary proteins are the source of essential amino acids.
- (4) Water is the most abundant chemical in living organisms.

109. The correct reason that explains why lipids are observed in acid-insoluble fraction obtained upon chemical analysis of a living tissue is that

- (1) Molecular weight of lipids are in the range of ten thousand Daltons and above
They are present in cell membrane and when these membranes are broken into pieces during grinding of a tissue, they form vesicles which are not water-soluble
- (2) They are heteropolymeric *i.e.*, made up of different repeating units
- (3) Their molecular weights do not exceed beyond 800 Da.

110. Exoskeleton of arthropods is mainly composed of a complex polysaccharide called 'X' whose repeating unit is 'Y'.

Choose the option that **correctly** identifies 'X' and 'Y' respectively.

- (1) Glucagon, Glucose
- (2) Insulin, Fructose
- (3) Chitin, N-acetyl glucosamine
- (4) Starch, N-acetyl galactosamine

111. Fibroblasts secrete fibres that provide all of the following properties to the connective tissues, **except**

- (1) Strength
- (2) Flexibility
- (3) Elasticity
- (4) Excitability

112. Select the **incorrect** statement.

Prevention or early detection and cure of STIs are given prime consideration under the reproductive health-care programmes.

Some of STIs like hepatitis-B can also be transmitted from infected person by sharing of injection needles and surgical instruments.

A large number of couples all over the world including India are infertile *i.e.*, they are unable to produce children inspite of unprotected sexual co-habitation.

In IUI, embryo with more than 8 blastomeres collected from healthy donor is artificially introduced either into the vagina or into the uterus.

113. In human males, the enlarged end of penis is called

- (1) Hymen
- (2) Urethral meatus
- (3) Glans penis
- (4) Foreskin

114. Choose the **correct** match w.r.t. humans.

(1)	Sertoli cells	–	Secrete androgens
(2)	Leydig cells	–	Secrete testosterone
(3)	Male germ cells	–	Present in interstitial spaces of testis
(4)	Immunocompetent cells	–	Present in lumen of seminiferous tubules

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

115. The protective coloration called mimicry is shown by

- (1) Male frogs
- (2) Male *Periplaneta*
- (3) Female *Periplaneta*
- (4) Male humans

116. Which of the following prokaryotic organisms can live in extremely salty areas?

- (1) Halophiles
- (2) Thermoacidophiles
- (3) Methanogens
- (4) Eubacteria

117. Mad cow disease in cattle is caused by an agent which has/is

- (1) Devoid of any biomolecule
- (2) Free DNA without protein coat
- (3) Abnormally folded protein
- (4) Free RNA without protein coat

118. Read the following statements and select the **correct** answer.

Statement A: Some epidermal cells from the zone proximal to the region of elongation towards the stem base form very fine and delicate thread-like structures called root hair.

Statement B: The root hair increases the surface area for the absorption of water and minerals from the soil.

- (1) Only statement A is correct
- (2) Only statement B is correct
- (3) Both the statements A and B are correct
- (4) Both the statements A and B are incorrect

- 119.** Match the column I with column II and select the **correct** option.

	Column I (Types of placentation)		Column II (Plant)
(a)	Marginal	(i)	<i>Dianthus</i>
(b)	Axile	(ii)	<i>Argemone</i>
(c)	Parietal	(iii)	Lemon
(d)	Free central	(iv)	Pea

- (1) a(iv), b(iii), c(i), d(ii)
 (2) a(iv), b(iii), c(ii), d(i)
 (3) a(iii), b(iv), c(i), d(ii)
 (4) a(ii), b(iv), c(i), d(iii)

- 120.** Identify the **incorrectly** matched pair.

(1)	Phyllodes	–	Australian acacia
(2)	Hypogynous flower	–	<i>Petunia</i>
(3)	Perigynous flower	–	Mustard
(4)	Epigynous flower	–	Cucumber

- (1) (1)
 (2) (2)
 (3) (3)
 (4) (4)

- 121.** Read the following statements (A – D).

(A) Cyanobacteria are photosynthetic prokaryotes, being the most primitive organism to have oxygenic photosynthesis.
 (B) Bacterial structure is very simple but they are very complex in their behaviour.
 (C) Smallest wall-less monerans are mostly autotrophic.
 (D) Archaeobacteria differ from other bacteria in having a different cell wall structure which is responsible for their survival in extreme conditions.

Select the option for the **correct** set of statements.

- (1) Only (A), (B) and (C)
 (2) Only (B) and (D)
 (3) Only (C) and (D)
 (4) Only (A), (B) and (D)

- 122.** Read the following Assertion (A) and Reason (R) and select the **correct** option.

Assertion (A): The pyramid of biomass in a sea is generally inverted.

Reason (R): Biomass of zooplanktons is higher than that of phytoplanktons as life span of former is longer and the latter multiply much faster though having shorter life span.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 (3) (A) is true statement but (R) is false
 (4) Both (A) and (R) are false

- 123.** Jan Ingenhousz through his experiment showed that

- (1) Glucose synthesized during photosynthesis is usually stored as starch
 (2) Only green parts of plant can synthesize glucose
 Sunlight is essential to the plant process that somehow
 (3) purifies the air fouled by burning candles or breathing animal
 (4) Carbon dioxide is essential for the synthesis of carbohydrates in plants

- 124.** Which among the following pigments shows highest peak in absorption spectrum between 400-500 nm wavelength of light?

- (1) Chlorophyll a
 (2) Chlorophyll b
 (3) Carotenoids
 (4) Xanthophylls

- 125.** What percent of energy in glucose is released in both alcoholic and lactic acid fermentation?

- (1) More than 36%
 (2) Less than 7%
 (3) About 13%
 (4) About 20%

- 126.** In which of the following conditions the rate of decomposition will be slower?

- (1) Presence of warm temperature
 (2) Presence of optimum soil moisture
 (3) Detritus rich in nitrogen
 (4) Under anaerobic condition

- 127.** Match the following columns and select the correct option w.r.t different complexes in ETS.

	Column I		Column II
(a)	Complex II	(i)	Cytochrome c oxidase
(b)	Complex IV	(ii)	Cytochrome bc_1 complex
(c)	Complex I	(iii)	Succinate dehydrogenase
(d)	Complex III	(iv)	NADH Dehydrogenase

- (1) a(iii), b(iv), c(i), d(ii)
 (2) a(iv), b(iii), c(ii), d(i)
 (3) a(iii), b(i), c(iv), d(ii)
 (4) a(iv), b(i), c(ii), d(iii)

- 128.** Pollen grain in gymnosperms represent

- (1) Reduced endosperm
 (2) Male cone
 (3) Microsporangium
 (4) Reduced male gametophyte

- 129.** Prothallus of *Dryopteris* is
- Inconspicuous
 - Unicellular
 - Diploid
 - Photosynthetic sporophyte
 - Free-living
 - Monoecious

- Only a, b and d
- Only c, d and e
- Only b, c and d
- Only a, e and f

- 130.** Match the columns and select the **correct** option.

	Column I		Column II
(a)	Toddy	(i)	<i>Penicillium</i>
(b)	Roquefort cheese	(ii)	<i>Lactobacillus</i>
(c)	Baker's yeast	(iii)	<i>Caryota urens</i>
(d)	Curd	(iv)	<i>Saccharomyces</i>

- a(i), b(iii), c(ii), d(iv)
- a(iii), b(i), c(iv), d(ii)
- a(ii), b(i), c(iii), d(iv)
- a(i), b(iii), c(iv), d(ii)

- 131.** In human females, the gametes as well as several steroidal hormones are secreted and released by the

- Primary sex organ
- Secondary sex organ
- Accessory sex gland
- External genitalia

- 132.** Choose the correct statement w.r.t. the morphology of *Periplaneta americana*.

- Head is triangular in shape and lies posteriorly at right angle to the longitudinal axis of the body.
- Head is formed by the fusion of ten embryonic segments in adults.
- Head capsule bears two pairs of compound eyes only.
- Hypopharynx lies within the cavity enclosed by the mouthparts.

- 133.** Match column I and column II w.r.t. humans.

	Column I		Column II
a.	Chorionic villi	(i)	Embryo with 8-16 blastomeres
b.	Trophoblast	(ii)	Surrounded by uterine tissue
c.	Inner cell mass	(iii)	Outer layer of blastocyst
d.	Morula	(iv)	Forms all organs and organ system of the foetus

Select the correct option.

- a(i), b(iii), c(ii), d(iv)
- a(ii), b(iii), c(iv), d(i)
- a(iv), b(iii), c(ii), d(i)
- a(iii), b(iv), c(i), d(ii)

- 134.** Genetically engineered plants

- Are made by transferring the gene of interest in their roots via retroviruses.
- Can help to prevent vitamin-A deficiency in humans
- Like 'pomato' was very successful at commercial scale.
- Are not able to utilise the minerals present in soil efficiently

- 135.** A collection of methods that allows correction of a gene defect by replacing an abnormal gene by a normal gene is

- Gene cloning
- Gene amplification
- Gene therapy
- Electrophoresis

- 136.** Match column I with column II and select the correct option.

	Column I		Column II
a.	<i>Pheretima</i>	(i)	Internal fertilisation
b.	<i>Anopheles</i>	(ii)	Water canal system
c.	<i>Spongilla</i>	(iii)	Terrestrial organism
		(iv)	Water vascular system

- a(i), b(ii), c(iv)
- a(iii), b(i), c(iv)
- a(iii), b(i), c(ii)
- a(iv), b(i), c(iii)

- 137.** Parapodia are lateral appendages present in

- Aquatic arthropods
- Aquatic annelids
- Terrestrial roundworms
- Aquatic flatworms

138.Assertion(A) : Few years prior to 1983, diabetic patients had to use insulin extracted from pancreas of slaughtered cattle and pigs.

Reason(R) : In 1983, an American company prepared two DNA sequences corresponding to A and B chains of human insulin and introduced them in plasmids of *Entamoeba coli* to produce insulin chains.

In the light of above statements, select the **correct** option

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) Both (A) and (R) are false
- (4) (A) is true but (R) is false

139.“Poikilotherms lack the capacity to regulate their body temperature.”

The above-mentioned statement holds true for

- a. *Hippocampus*
- b. *Hyla*
- c. *Chelone*
- d. *Bangarus*

Choose the option with the **correct** set only.

- (1) Only a and c
- (2) Only b and d
- (3) Only a, b and c
- (4) a, b, c and d

140.During the course of evolution, invertebrates were formed and active by the time of

- (1) 1000 mya
- (2) 600 mya
- (3) 800 mya
- (4) 500 mya

141.In the course of animal evolution, thecodonts were descendents of

- (1) Pelycosaurs
- (2) Sauropsids
- (3) Therapsids
- (4) Synapsids

142.How many of the plants mentioned in the box below have directly evolved from Psilophyton?

Horsetails, Ginkgos, Ferns, Conifers, Seed ferns, Cycads

Choose the **correct** answer from the options given below.

- (1) Six
- (2) Four
- (3) Five
- (4) Three

143.In humans, which of the following events does not involve in micturition?

- (1) Stretching of the urinary bladder
- (2) Contraction of smooth muscles of urinary bladder
- (3) Voluntary signals by central nervous system
- (4) Contraction of urethral sphincter

144.Modern *Homo sapiens* arose

- (1) Near east and central Asia around 2 mya
- (2) 1.5 million years ago with a brain size of 900cc
- (3) During ice age between 75,000-10,000 years ago
- (4) Only in Tanzania around 15 mya and were probably not taller than 4 feet

145.Read the following statements A and B.

Statement A: Collecting duct plays a role in the maintenance of pH and ionic balance of the body by selective absorption of H^+ and various ions from the filtrate.

Statement B: Collecting duct allows passage of small amounts of urea into the medullary interstitium to maintain the osmolarity.

Select the **correct** answer from the options given below.

- (1) Both statements A and B are correct
- (2) Only statement A is incorrect
- (3) Only statement B is incorrect
- (4) Both statements A and B are incorrect

146.Assertion (A): Both rheumatoid arthritis and muscular dystrophy are auto-immune disorders.

Reason (R): Our body is capable of distinguishing between foreign molecules/organisms and self-cells.

In the light of above statements, select the **correct** option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) (A) is false but (R) is true

147.Consider the following:

- a. Influences libido
 - b. Produces anabolic effect on protein metabolism
 - c. Produces synthetic effect on carbohydrate metabolism
 - d. Regulates development and maturation of prostate gland
- Which of the above given functions are true for testosterone?

- (1) b and c only
- (2) a and d only
- (3) a, b and c only
- (4) a, b, c and d

148. Select the **incorrect** match w.r.t. gland and its location.

(1)	Parathyroid gland	–	One pair each in two lobes of the thyroid gland
(2)	Pineal gland	–	Dorsal side of forebrain
(3)	Thyroid gland	–	Either side of trachea interconnected by isthmus
(4)	Thymus	–	Behind sternum on dorsal side of aorta

- (1) (1)
(2) (2)
(3) (3)
(4) (4)

149. The causative agent of ringworms is _____ which can thrive in skin folds such as groin.
Choose the **correct** option to fill in the blank.

- (1) *Wuchereria*
(2) *Trichophyton*
(3) *Toxoplasma*
(4) *Ancylostoma*

150. Which of the following is not one of the measures taken for prevention and control of alcohol and drug abuse among adolescents?

- (1) Not pushing child unduly to perform beyond his/her threshold limits
(2) Seeking help only from parents and never from peers for guidance
(3) Help or counselling from highly qualified psychologists, psychiatrists and de-addiction and rehabilitation programmes
(4) Educating adolescents to face problems and to deal with disappointments in a positive way.

151. The oxygen dissociation curve shifts to the right side when there is

- (1) Increase in $p\text{CO}_2$
(2) Increase in pH
(3) Decrease in H^+ concentration
(4) Increase in $p\text{O}_2$

152. Read the following statements w.r.t. an adult human skeleton system.

- a. Fibrous joint is present between the flat skull bones
b. Fibrous cartilage is present between two halves of the pelvic girdle
c. In between atlas and axis, cartilaginous joint is present.
Choose the **correct** option.

- (1) a, b and c are correct
(2) Only a and b are correct
(3) Only b and c are correct
(4) Only c is correct

153. The respiratory disorder emphysema is attributed to

- (1) Lung fibrosis
(2) Increased activity of mast cells in the muscular surface of lungs
(3) Reduced respiratory surface area
(4) Inflammation of only bronchioles

154. What will be the heart rate per minute of a healthy adult person, if his cardiac output is 5000 mL and stroke volume is 50 mL?

- (1) 100
(2) 150
(3) 75
(4) 72

155. Myoglobin differs from haemoglobin as

- (1) Latter exhibits quaternary structure of proteins
(2) Latter does not help in transport of respiratory gases
(3) Former is absent in red muscle fibres
(4) Latter is not an iron containing pigment

156. Biological control methods adopted in agricultural pest control are based on the ability of the

- (1) Predator to achieve very high population densities
(2) Exotic species to increase the productivity of the invaded ecosystem
(3) Predator to maintain the species diversity in a community by increasing the intraspecific competition between prey species
(4) Predator to keep prey population under control without overexploiting its prey

157. Which of the following are the characteristics of a stable community?

- (i) Show too much variation in productivity year after year
(ii) Resistant to occasional disturbances
(iii) It has more species
(iv) Resistant to invasion by alien species

- (1) Only (i), (ii) and (iii)
(2) Only (ii), (iii) and (iv)
(3) Only (ii) and (iv)
(4) All (i), (ii), (iii) and (iv)

158. The IUCN Red list (2004) documents important recent extinction of Thylacine that was native to

- (1) Russia
(2) Australia
(3) Mauritius
(4) Bali

159.Read the following five statements (A to E) and select the option with all the **correct** statements.

- (A) The plant body of bryophytes is more differentiated than that of algae.
 (B) Gemmae are green, multicellular, asexual buds, which develop in small receptacles called gemma cups located on the thalli of mosses.
 (C) Gymnosperms can be found in extreme temperature areas.
 (D) In gymnosperms, one of the megaspores enclosed within the megasporangium develops into a multicellular female gametophyte that bears two or more female sex organ.
 (E) In *Equisetum* and *Selaginella*, sporophylls aggregate to form distinct male and female strobili or cones.

- (1) Only (A), (B) and (C)
 (2) Only (B), (C) and (E)
 (3) Only (C), (D) and (E)
 (4) Only (A), (C) and (D)

160.The very familiar beetle with red and black marking are useful to get rid of

- (1) Aphids
 (2) Mosquitoes
 (3) Butterfly caterpillars
 (4) Dragonflies

161.The monarch butterfly acquires a chemical highly distasteful to its predator (bird) during its

- (1) Adult stage
 (2) Caterpillar stage from its parents
 (3) Caterpillar stage by feeding on a poisonous weed
 (4) Zygote stage

162.Heterophyllous development due to environment can be observed in

- (1) Larkspur
 (2) Coriander
 (3) Buttercup
 (4) Cotton

163.Choose the **incorrect** statement from the following.

- (1) Growth is an irreversible increase, expressed in various parameters
 (2) All plants show primary and secondary growth
 (3) A differentiated cell may dedifferentiate and then redifferentiate
 (4) Plants retain capacity of unlimited growth throughout the life.

164.Read the following Assertion (A) and Reason (R) statements and choose the **correct** option.

Assertion (A): Cultivation of hybrids has tremendously increased productivity but hybrid seeds have to be produced every year.

Reason (R): When seeds from the hybrids are sown, the plants in the progeny lose the hybrid characters.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
 (3) (A) is true but (R) is false
 (4) Both (A) and (R) are false

165.Which of the following is a parasitic plant that contains very large number of seeds?

- (1) *Ficus*
 (2) *Orobancha*
 (3) *Orchid*
 (4) *Lupinus*

166.Read the statements given below and identify the **correct** option.

Statement A: Endosperm development precedes embryo development.

Statement B: Early stages of embryogeny in monocots differs from that of dicots.

- (1) Both the statement A and statement B are true
 (2) Both the statement A and statement B are false
 (3) Statement A is true but statement B is false
 (4) Statement B is true but statement A is false

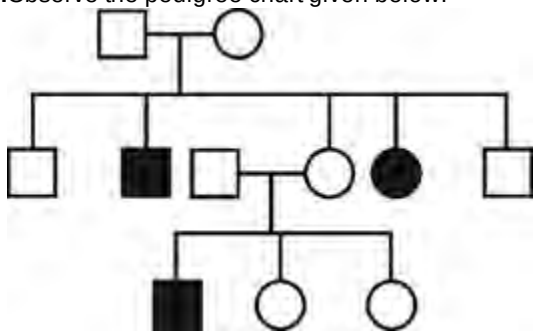
167.How many type of gametes can be produced by a diploid organism, if it is heterozygous for 4 loci?

- (1) 8
 (2) 4
 (3) 16
 (4) 32

168.Which of the following is **not** true w.r.t. the law of independent assortment?

- (1) It cannot be explained by using a monohybrid cross
 (2) This law is not universally applicable
 (3) Closely located genes always assort independently
 (4) It is based on generalisations proposed by Mendel

169. Observe the pedigree chart given below:



The above pedigree chart **cannot** be applied for

- (1) Myotonic dystrophy
- (2) Sickle-cell anaemia
- (3) Phenylketonuria
- (4) Cystic fibrosis

170. Identify the **incorrect** statement for polygenic inheritance.

- (1) All the alleles show additive effect.
- (2) It takes into account, the influence of environment besides the involvement of multiple genes.
- (3) These traits are generally controlled by two or more alleles of a gene.
- (4) Kernel colour in wheat is an example of polygenic inheritance.

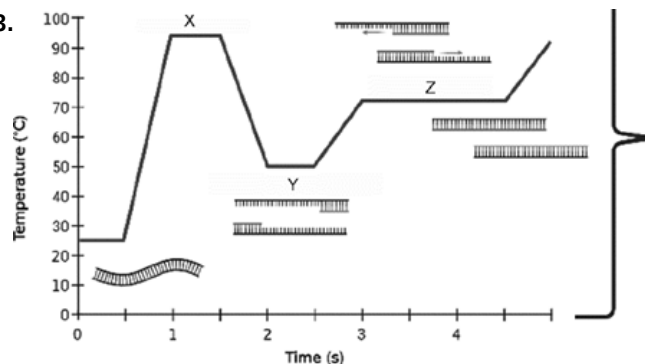
171. Which of the following is **true** for white muscle fibres as compared to red muscle fibres?

- (1) Contain large number of mitochondria
- (2) Have low amount of sarcoplasmic reticulum
- (3) Depend on anaerobic process for energy
- (4) Can store large amount of oxygen

172. In a standard ECG, the end of ventricular systole is marked by

- (1) P-wave
- (2) QRS complex
- (3) Start of T-wave
- (4) End of T-wave

173.



The figure given above represents the variations in temperature for a PCR set up and the given labellings represent the different steps involved in it.

Choose the correct option w.r.t. the given figure.

- (1) 'Y' occurs at around 60° C when primers attach to the 5' ends of the template strands
- (2) 'Z' occurs in the presence of a heat labile DNA polymerase
- (3) 'Y' and 'Z' occur at high temperature which lead to the breakdown of phosphodiester bonds between the template strands and the primer
- (4) 'X' occurs at high temperature to generate two separate template DNA strands

174. If a foreign gene is inserted at *Pvu* I site of pBR322 followed by its insertion in host cells, then

- (1) The recombinants will lose resistance against tetracycline
- (2) All transformants will be both ampicillin and tetracycline resistant
- (3) The non-recombinant transformants will be resistant to both ampicillin and tetracycline
- (4) The recombinants will grow on ampicillin containing medium

175. Select the part of the human brain which has centres that can moderate the cardiac functions through the autonomic nervous system.

- (1) Cerebrum
- (2) Midbrain
- (3) Medulla oblongata
- (4) Corpora quadrigemina

176. During the depolarisation of a neuron, 'X' moves inside the intracellular fluid of axon by voltage-gated channels while during repolarisation of the neuron, 'Y' moves outside from intracellular fluid of axon by voltage-gated channels. Identify 'X' and 'Y' respectively and choose the **correct** option.

- (1) K^+ , Na^+
- (2) Ca^{2+} , Na^+
- (3) Na^+ , K^+
- (4) K^+ , Ca^{+2}

177.For transformation, the micro-particles coated with DNA are bombarded with gene gun. These micro-particles are made up of

- (1) Gold or Tungsten
- (2) Silicon or Iron
- (3) Steel or Zinc
- (4) Silicon or Diamond

178.Choose the **odd** one w.r.t. features of a plasmid.

- (1) Autonomous replication
- (2) Transferable
- (3) Always linear
- (4) Extra-chromosomal DNA

179.Which of the following is used as the staining dye during gel electrophoresis that gives orange-coloured DNA bands when observed under the UV light as it stacks between the base pairs of DNA?

- (1) Ethidium bromide
- (2) Chilled ethanol
- (3) Bromophenol blue
- (4) Trichloroacetic acid

180.Which of the following have similar mode of action in terms of preventing conception?

- (1) Progestogens injections and steroidal oral contraceptive pills
- (2) Implants containing oxytocin and non-steroidal oral contraceptive pills
- (3) Barrier contraceptives and non-steroidal oral contraceptive pills
- (4) Terminal method of contraception and chemical contraceptives